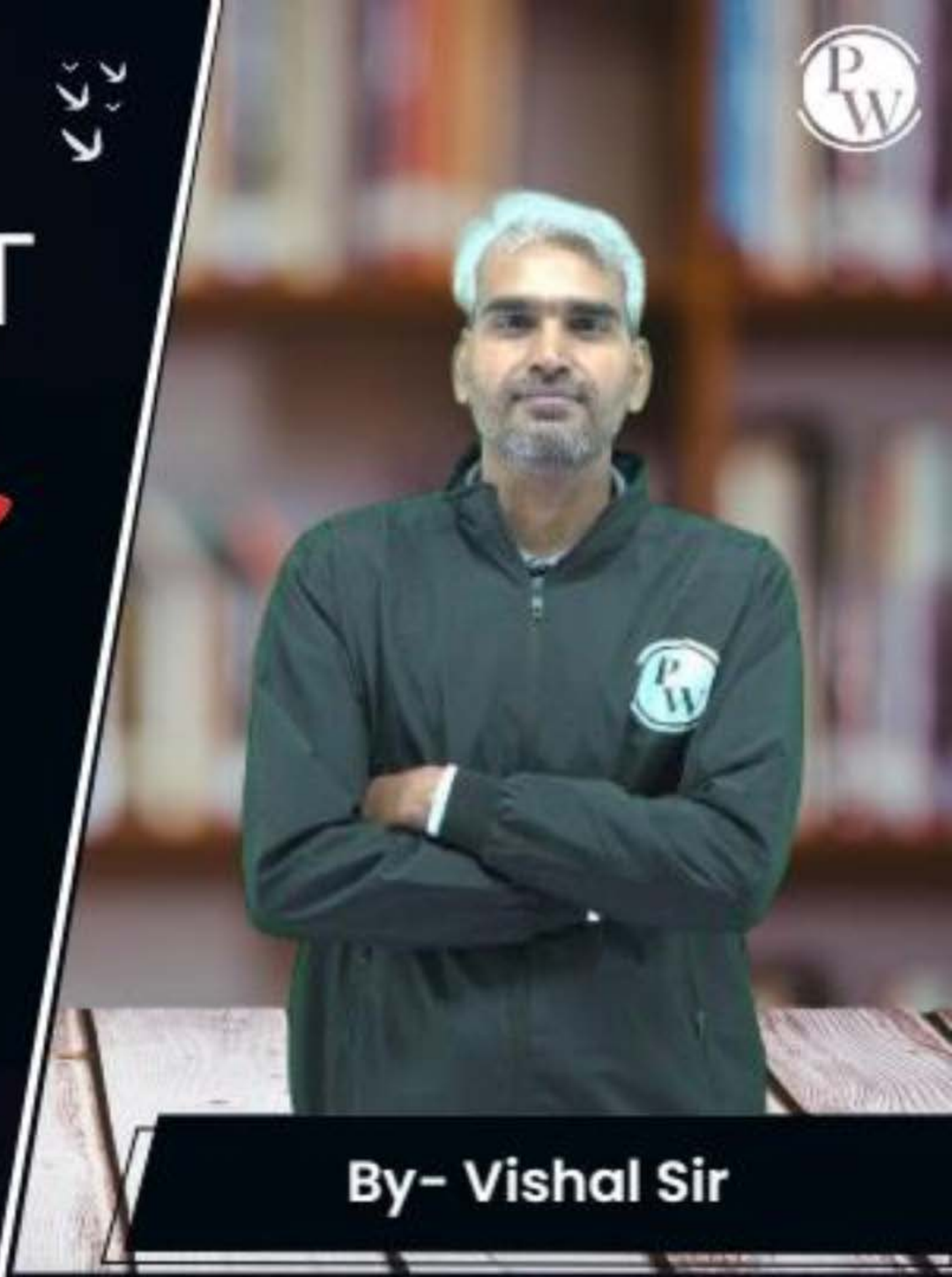
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 20



By- Vishal Sir

Recap of Previous Lecture



Topic Number of onto functions

Topic Bijective function & Inverse of a function

Slide

Topics to be Covered



Topic — Identity function and constant function

Topic — Identical function

Topic — Function composition

Slide

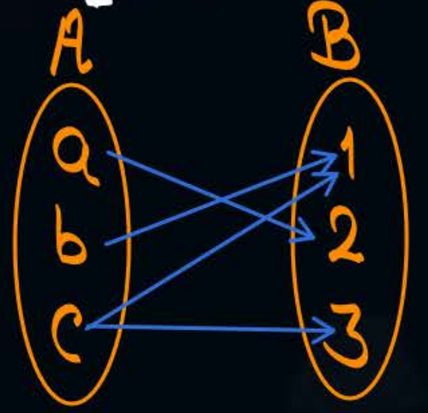


Topic : Function

Every function from set A to set B is also a relation from A to B, but Every relation from set A to set B need not be a function from A to B

- A relation from set A to set B is called a function from set A to set B if and only if Every element of set A relates with exactly '1' element of set B.

eg: $R_1 : A \to B$ { R_1 is a relation from A to B }



{ element 'c' of set A relates with two elements of set B, ∴ Not a function }

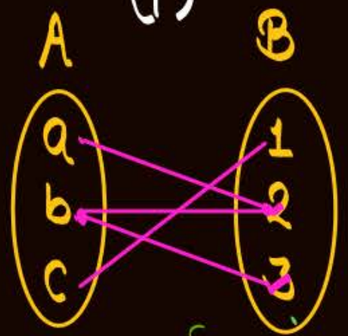
eg: $R_2 : A \to B$ { R_2 is a relation from A to B }



{ Element 'b' relates with '0' element of set B, ∴ Not a function }

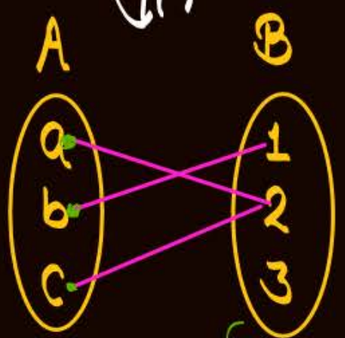
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(i)



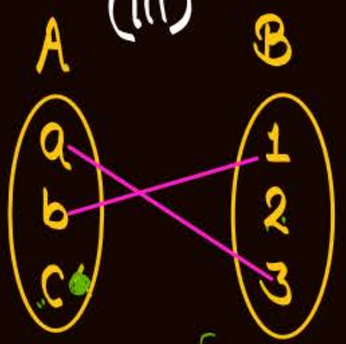
Neither a function from A to B, nor a function from B to A

(ii)



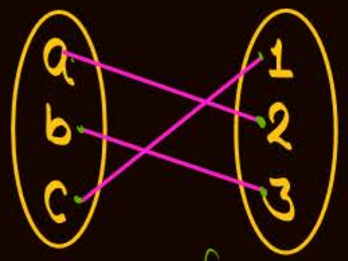
It is a function from A to B, but not a function from B to A

(iii)



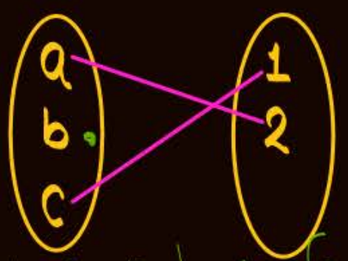
Neither a function from A to B, nor a function from B to A

(iv)



It is a function from A to B as well as a function from B to A

(v)



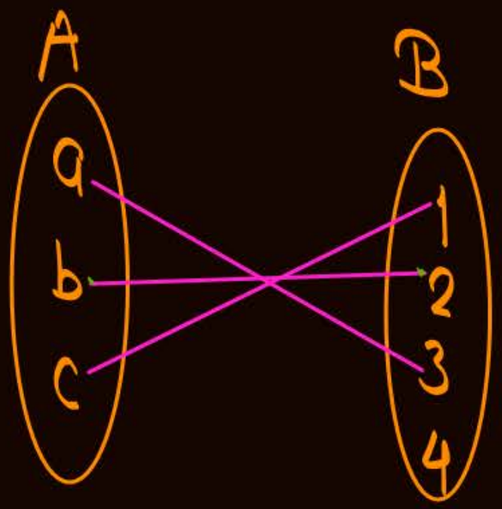
It is not a function from A to B, but a function from B to A

- Let $f: A \to B$ is a function,

then set 'A' is called domain of the function 'f'

and set 'B' is called Co-domain of the function 'f'

Eg: $f: A \to B$



domain of f = {a, b, c}

Co-domain of f = {1, 2, 3, 4}

Here $f(b) = 2$,

i.e. image of element 'b' w.r.t. function 'f' is '2'.

(or)

We say that Pre-image of '2' w.r.t. function 'f' is 'b'.



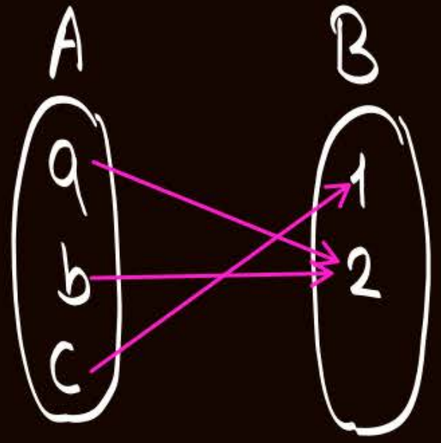
Topic : Range of a function

The set of all elements from Co-domain that are mapped by at least one element of domain is called "Range" of the function.

Range of function $f \subseteq$ Co-domain of function f

Slide

eg: $f_1 : A \to B$



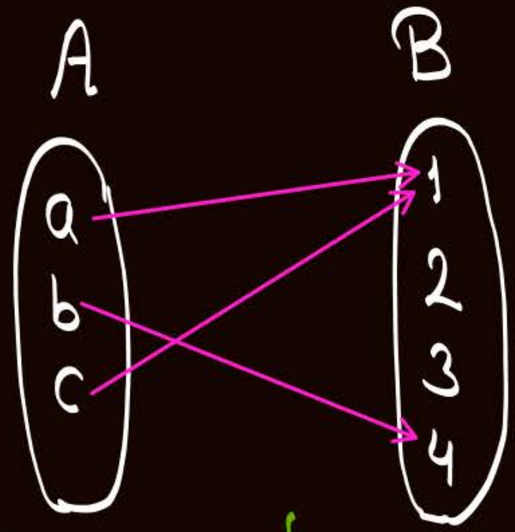
Domain : {a, b, c}

Co-domain : {1, 2}

Range = {1, 2}

Range = Co-domain

$f_2 : A \longrightarrow B$



Domain : {a, b, c}

Co-domain = {1, 2, 3, 4}

Range = {1, 4}

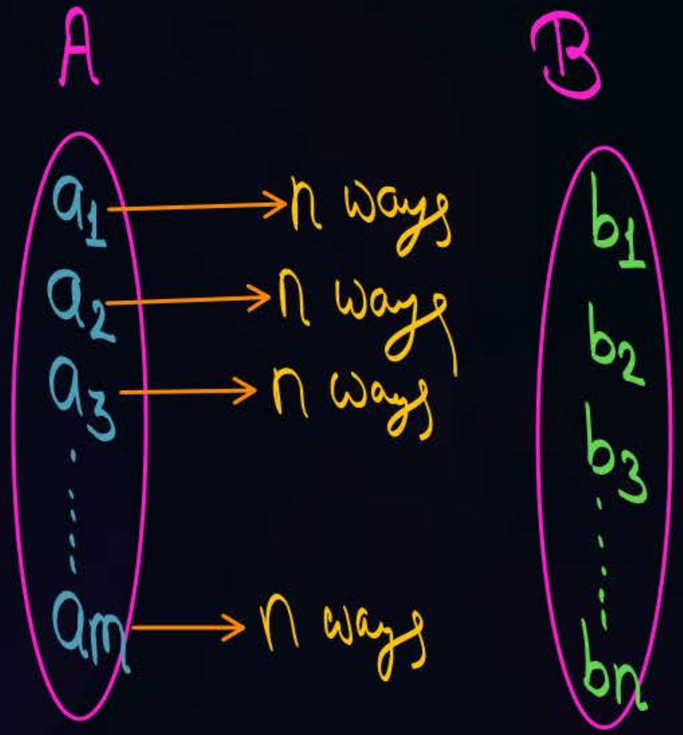
Range $\subset$ Co-domain



Topic : Total number of functions

* Let $|A| = m$ and $|B| = n$, then

Total Number of Functions Possible from set A to set B $= n^{m}$ $= (\text{Size of Codomain})^{\text{Size of domain}}$



∴ Total no. of functions Possible from set A to set B $= \underbrace{n * n * \cdots * n}_{m \text{ times } n}$

$= n^{m}$

Slide

Note:- A function defined from set A to set A itself is called a function on set A.

If $|A| = n$, then

Total no. of functions possible on set A (from A to A) $= (\text{Size of Codomain})^{\text{Size of domain}} = (|A|)^{|A|} = n^n$



Topic : Injective (one-one) function

A function $f: A \to B$ is called an injective (one-one) function if and only if:

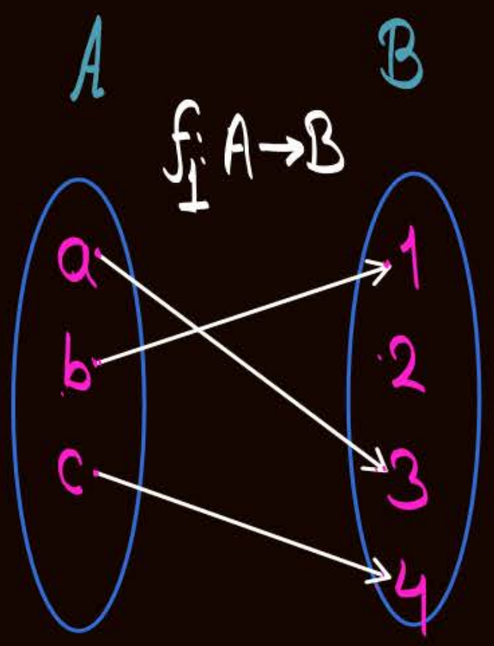
* distinct elements of the domain (set A) have distinct images in codomain (set B)

i.e. $\forall\, a, b \in$ domain

if $a \neq b$ then $f(a) \neq f(b)$

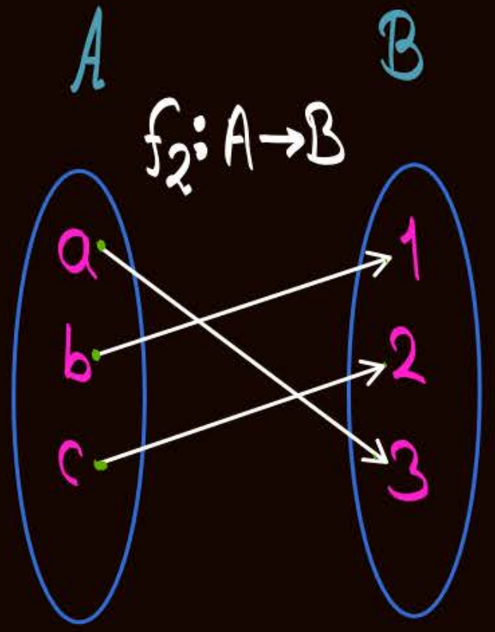
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$|A| < |B|$



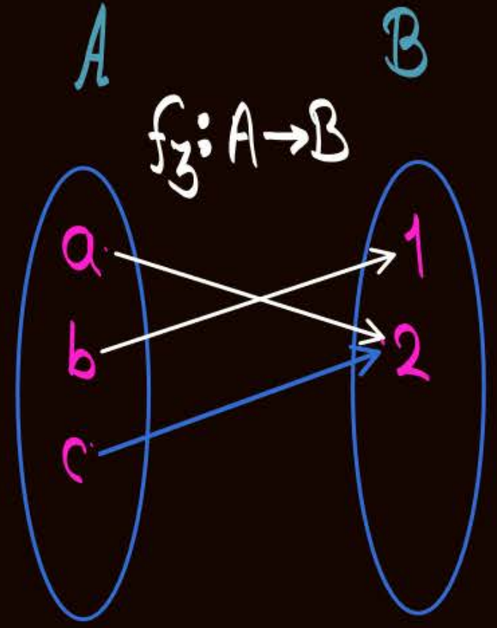
'f_1' is a One-one function from A to B

$|A| = |B|$



'f_1' is a One-one function from A to B

$|A| > |B|$



$a \neq c$ but $f(a) = f(c)$

$\therefore$ Not a one-one function

'f_3' is not one

Note:-

① If $|A| > |B|$, then
One-one function is not possible from set A to set B

② If One-one function exists from set A to set B, then $|A| \leq |B|$

③ In one-one function Every element of Co-domain can have at most '1' pre-image



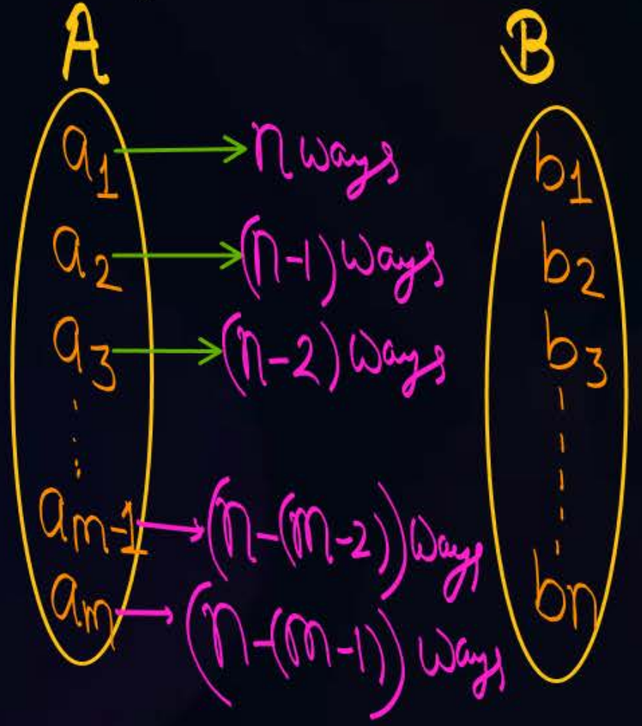
Topic : Number of one-one function

Let $|A| = m$ & $|B| = n$ (such that $m \leq n$),

then,

Numer of one-one functions possible from set A to set B $= \frac{n!}{(n-m)!} = {}^{n}P_{m}$

$\because {}^{n}P_{r} = \frac{n!}{(n-r)!}$



Number of one-one functions possible from set A to set B $= n*(n-1)*(n-2)*\cdots*(n-m+2)(n-m+1)$

$= n(n-1)(n-2)\cdots(n-m+1) * \frac{(n-m)(n-m-1)\cdots 3\cdot 2\cdot 1}{(n-m)(n-m-1)\cdots 3\cdot 2\cdot 1}$

$= \frac{n!}{(n-m)!}$

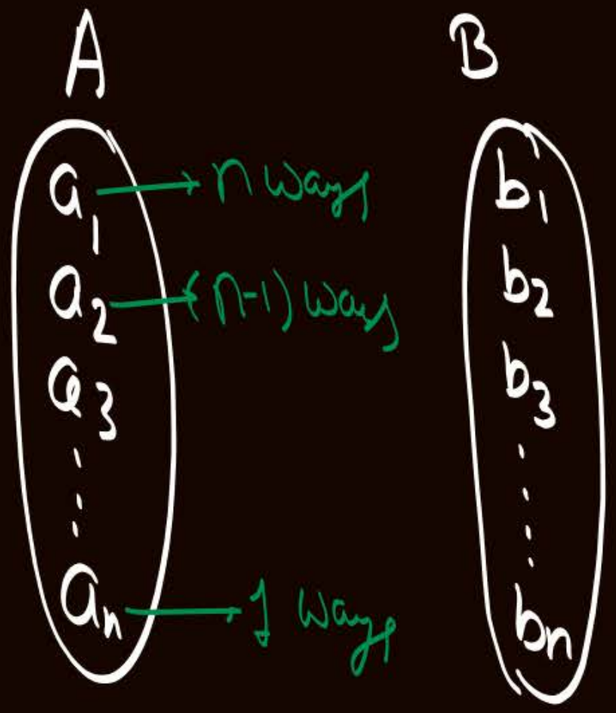
$\therefore$

Slide

Note: If $|A| = |B| = n$,

then, number of one-one function Possible from set A to set B $= {}^{n}P_{n} = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$

$0! = 1$



No. of one-one function Possible $= n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$

$= n!$



Topic : Surjective (onto) function

A function $f: A \to B$ is called an Onto (Surjective) function if and only if

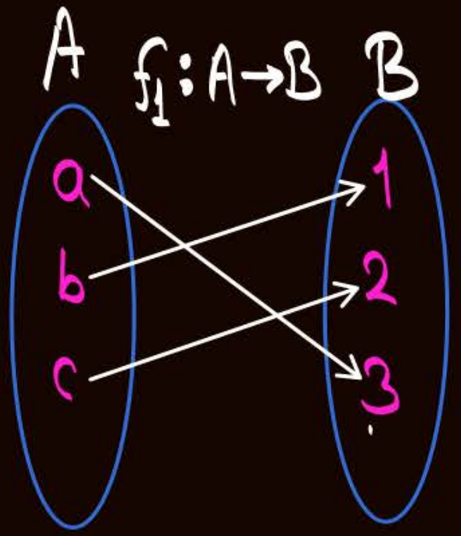
Every element of Co-domain is mapped by at least one element of domain

i.e. in onto function.

Range = Co-domain

Slide

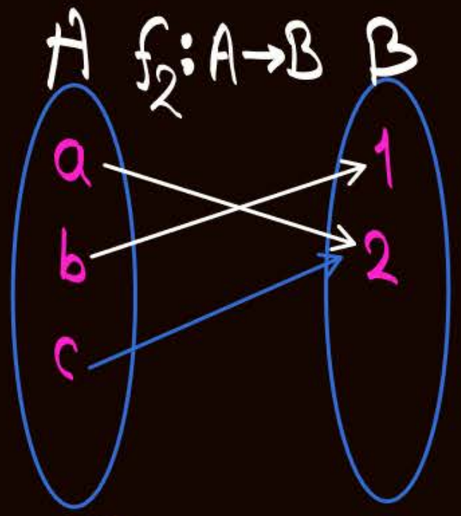
$|A| = |B|$



Range = Co-domain

$\therefore f_1$ is an onto function

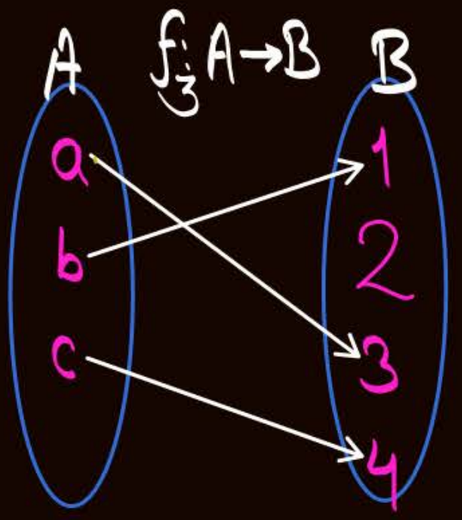
$|A| > |B|$



Range = Co-domain

$\therefore f_2$ is an onto function

$|A| < |B|$



Range $\subset$ Co-domain

$\therefore f_3$ is not an onto function

Note:

① If $|A| < |B|$, then Onto function is not possible from set A to set B.

② If there exist an onto function from set A to set B, then $|A| \geq |B|$

③ In an onto function Every element of Co-domain will have at least '1' pre-image



Topic : Number of onto functions

① If $|A| = |B| = n$, then
How many Onto functions are possible from A to B

② If $|A| = n$ & $|B| = 2$ such that $n \geq 2$, then
How many onto functions are possible from A to B

③ if $|A| = 4$ & $|B| = 3$, then
How many onto functions are possible from A to B.

Slide

H.W.

Q:- If $|A| = |B| = n$ then How many onto functions are possible from A to B.

Note:- If $|A| = |B|$, then
Every one-one function from set A to set B is onto,
and Every onto function from set A to set B is one-one

∴ If $|A| = |B|$, then
No. of onto function = No. of one-one function

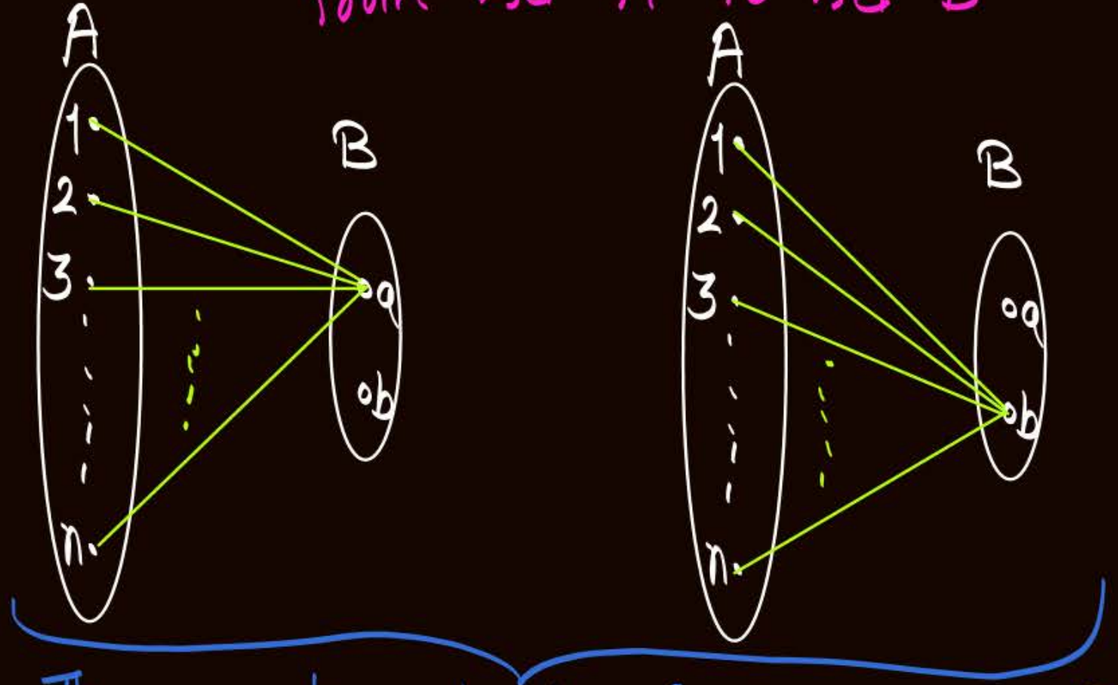
If $|A| = |B| = n$, then

No. of onto functions from set A to set B = No. of one-one functions from set A to set B = $n!$

H.W.

Q:- If $|A|=n$ and $|B|=2$ $(\text{s.t. } n \geq 2)$ then How many onto functions are possible from A to B.

No. of onto functions possible from set A to set B = (Total no. of functions Possible from A to B) − (No. of functions from A to B that are not onto)



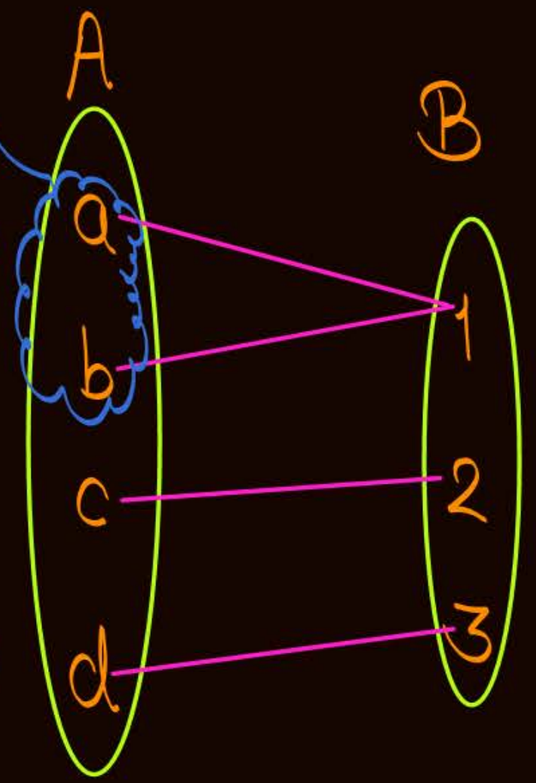
These are the only two functions possible from Set A to Set B that are not onto

$= (2)^n - 2$

$= \boxed{2^n - 2}$

H.W. Q:- If $|A|=4$ and $|B|=3$ then How many onto functions are possible from A to B.

Exactly two elements of set A will relate with same element of set B & other two elements of set A will relate with other two elements of set B.



No. of onto functions possible from set A to set B $= {}^{4}C_{2} * 3!$

$= 6 * 6$

$= \boxed{36}$

H.W.

Q:- If $|A| = n$ and $|B| = n-1$
then How many onto functions are possible from A to B.

No. of onto functions possible from set A to set B $= {}^{n}C_{2} * (n-1)!$

H.W.

Q: if $|A| = m$ & $|B| = n$ (such that $m \geq n$) then how many onto functions are possible from set A to set B.

No. of onto functions from set A to set B

$$= n^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 (n-3)^m + {}^nC_4 (n-4)^m - \cdots + (-1)^{n-1} {}^nC_{n-1} (n-(n-1))^m$$

$$= {}^nC_0 (n-0)^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 (n-3)^m + \cdots + (-1)^{n-1} {}^nC_{n-1} (n-(n-1))^m$$

$$= \boxed{\sum_{i=0}^{(n-1)} (-1)^i \, {}^nC_i \, (n-i)^m}$$

		No. of onto functions from set A to set B
①	$\lvert A\rvert = n$ & $\lvert B\rvert = 2$ $(n \geq 2)$	$2^n - 2$
②	$\lvert A\rvert = n$ & $\lvert B\rvert = n$	$n!$
③	$\lvert A\rvert = n$ & $\lvert B\rvert = n-1$ $\lvert A\rvert = 1 + \lvert B\rvert$	${}^{n}C_2 * (n-1)!$
④	$\lvert A\rvert = m$ & $\lvert B\rvert = n$ where $(m \geq n)$	$\sum_{i=0}^{(n-1)} (-1)^i \cdot {}^{n}C_i \cdot (n-i)^m$



Topic : Bijective Function

A function from set A to set B is called a bijective function if and only if (i) 'f' is one-one and (ii) 'f' is onto

$f: A \to B$ is bijective iff $f: A \to B$ is one-one & $f: A \to B$ is onto

i.e. $|A| \leq |B|$ & $|A| \geq |B|$

i.e. $f: A \to B$ is bijective then $|A| = |B|$

If $|A| = |B| = n$, then

No. of bijections possible from set A to set B $= n!$

Slide

Note :- ① If bijective function exist between the sets A and B, then Cardinality of set A = Cardinality of set B.

② If bijective function exist between the sets A and B, then set A and set B are said to have

One-one Correspondance.

(different from one-one function)

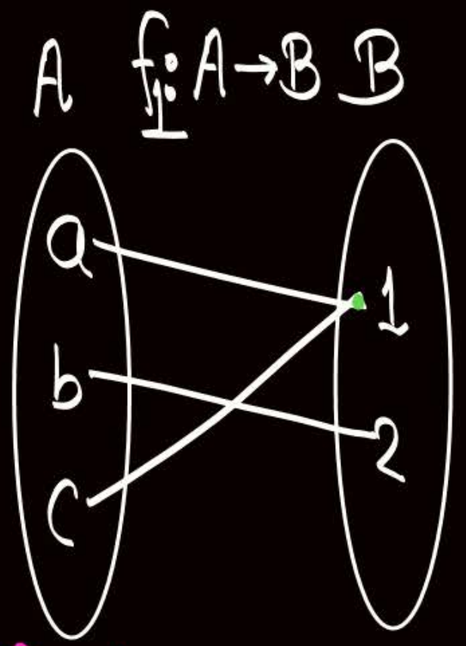


Topic : Inverse of a Function

- Inverse of a function "f" can exist if and only if function "f" is a bijective function

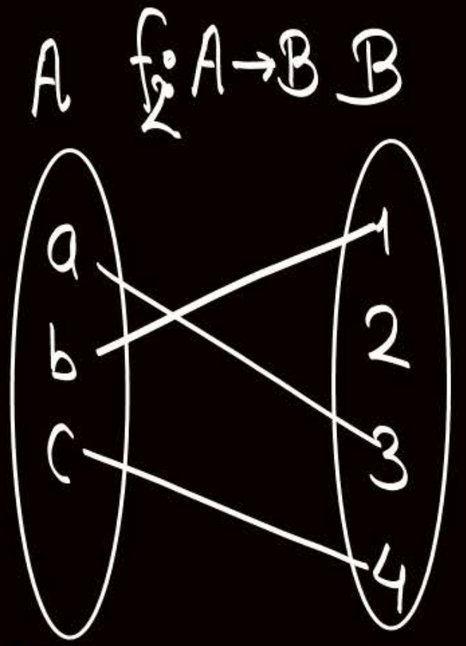
If $f: A \rightarrow B$ is a bijection, then inverse function of function 'f' will exist and it can be denoted by 'f^{-1}' and 'f^{-1}' will be a function from set B to set A.

Slide



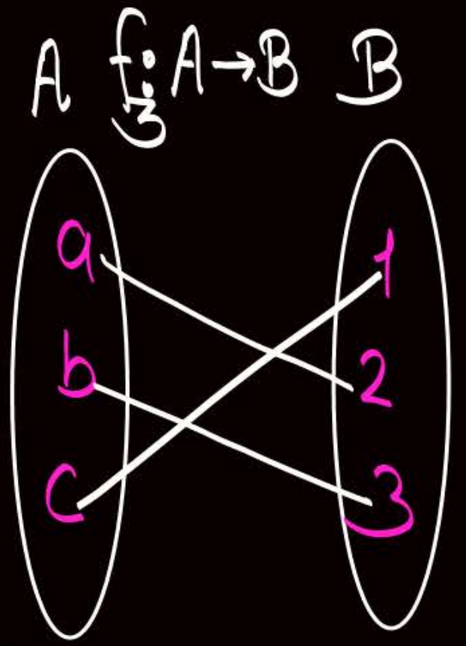
f_1 is onto, but not one-one

* Inverse relation wrt. function f_1 will not be a function



f_2 is one-one but not onto

* Inverse relation wrt. function f_2 will not be a function



f_3 is one-one as well as onto

* Inverse relation wrt. function f_3, will be a function

R :- Set of all real no.s.

N :- Set of all natural no.s

} Both are infinite set.

↳ We can not define a bijective function b/w R and N

⇓

i.e. Set of all real no.s is not equivalent to set of all Natural no.s.

S_1 = Set of all non-negative Even integers.

i.e. $S_1 = \{0, 2, 4, 6, 8, 10, \ldots\ldots\ldots\}$

&

S_2 = Set of all whole numbers.

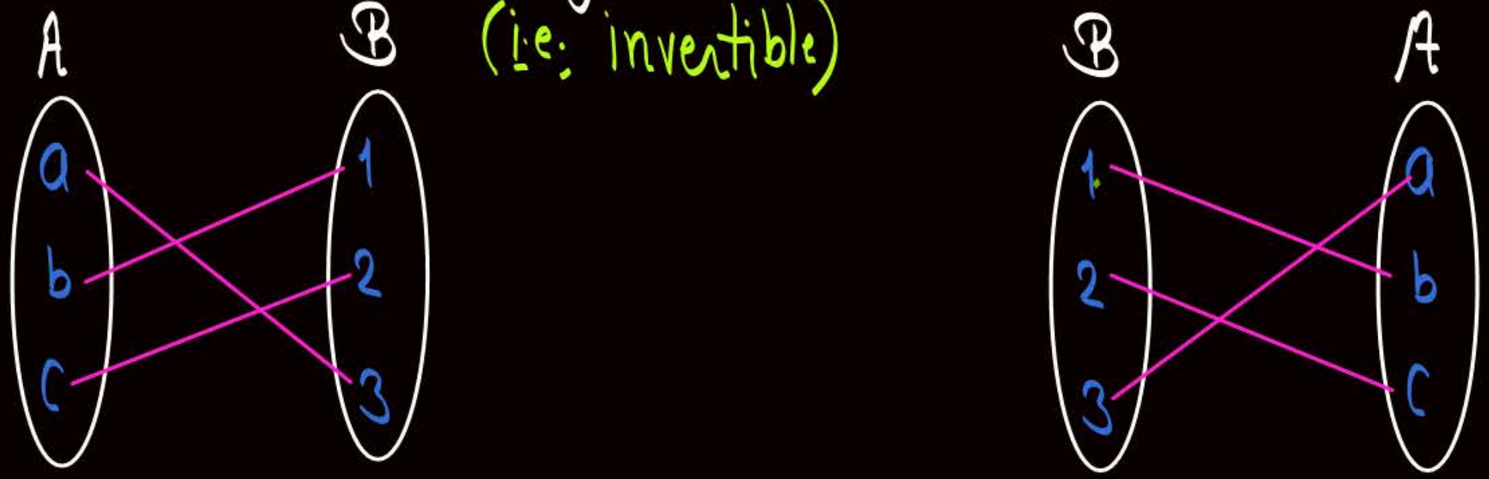
i.e. $S_2 = \{0, 1, 2, 3, 4, 5, \ldots\ldots\ldots\}$

Let $f: S_1 \to S_2$

s.t. $f(x) = \frac{x}{2}$

} it is a bijective function from S_1 to S_2
$\therefore$ S_1 is Equivalent to S_2

eg: let $f: A \to B$ is bijection $\therefore$ $f^{-1}: B \to A$ (i.e: invertible)



$f(a) = 3 \longrightarrow f^{-1}(3) = a$

$f(b) = 1 \longrightarrow f^{-1}(1) = b$

$f(c) = 2 \longrightarrow f^{-1}(2) = c$

i.e: if $f(x) = y$ then $f^{-1}(y) = x$

Q: Let 'f' is a bijective function defined as $f(x) = 2x+3$

Obtain the inverse function w.r.t. function 'f'

i.e., $f^{-1}(x) = ?$

Let $f(x) = y = 2x+3 \Longrightarrow y = 2x+3$

$\therefore x = \left(\frac{y-3}{2}\right)$

$\therefore f^{-1}(y) = x$

↳ $f^{-1}(y)$ is a function of 'y'

$\therefore$ We must represent 'x' in terms of y

Hence $f^{-1}(y) = \left(\frac{y-3}{2}\right)$

$\therefore \boxed{f^{-1}(x) = \frac{x-3}{2}}$

Today's Topic

H.W:

Q:

Gate 1996 :-

If $f(x,y) = (x+y, x-y)$ defined from $R \times R$ to $R \times R$.

Find $f^{-1}(x,y)$

Let $f(x,y) = (a,b) = (x+y, x-y)$ $\therefore$ $a = x+y$ — ①

& $b = x-y$ — ②

$\therefore f^{-1}(a,b) = (x,y)$

By eq^n ① & eq^n ②

$x = \frac{a+b}{2}$ & $y = \frac{a-b}{2}$

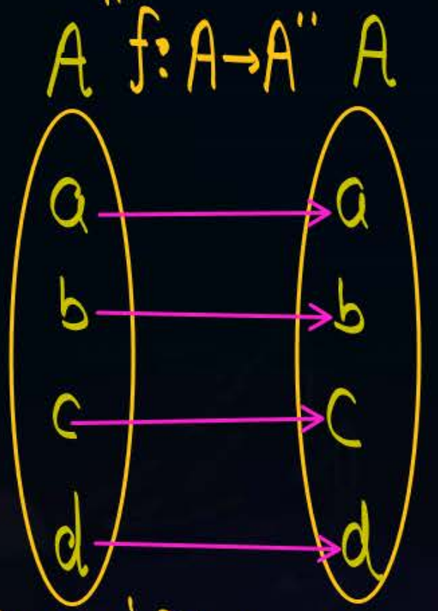
$f^{-1}(a,b) = \left(\frac{a+b}{2}, \frac{a-b}{2}\right)$

$\therefore f^{-1}((x,y)) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$



Topic : Identity Function

A function "f" from set A to set A (i.e. on set A) is called an identity function if and only if $f(x) = x, \forall x \in A$



Here 'f' is an identity function

- Generally, Identity function on set A is denoted by I_A.
- Every identity function is always a bijection
- Inverse of identity function will always be that identity function itself i.e $I_A^{-1} = I_A$

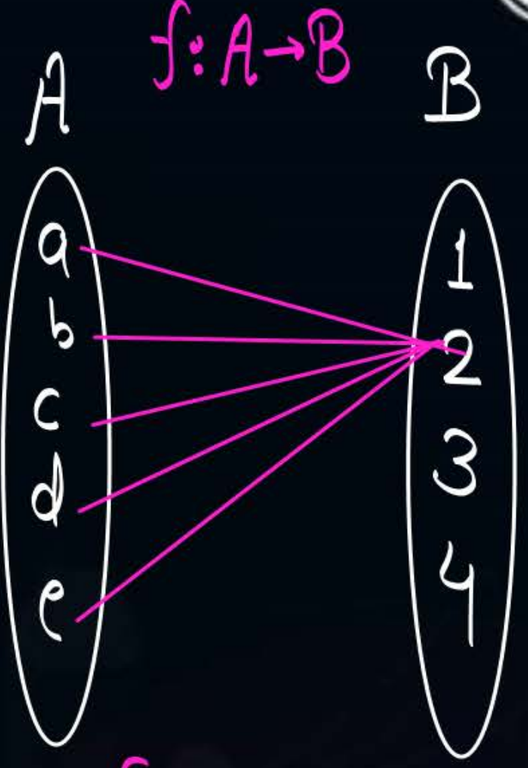
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Topic : Constant Function

A function $f: A \to B$ is called a Constant function, if and only if, $f(x) = C, \forall x \in A$

C ← fixed value from Co-domain



$f(x) = 2, \forall x \in A$

$\therefore f: A \to B$ is a Constant function

- If $|A| = m$ and $|B| = n$, then

No. of Constant functions possible from set A to set B $= n =$ Cardinality of Co-domain

Slide



Topic : Identical function / Equal functions

Two functions 'f' and 'g' are said to be identical if and only if

① Domain of f = Domain of g

and ② Range of f = Range of g

and ③ $f(x) = g(x) \quad \forall x \in$ Domain

Slide

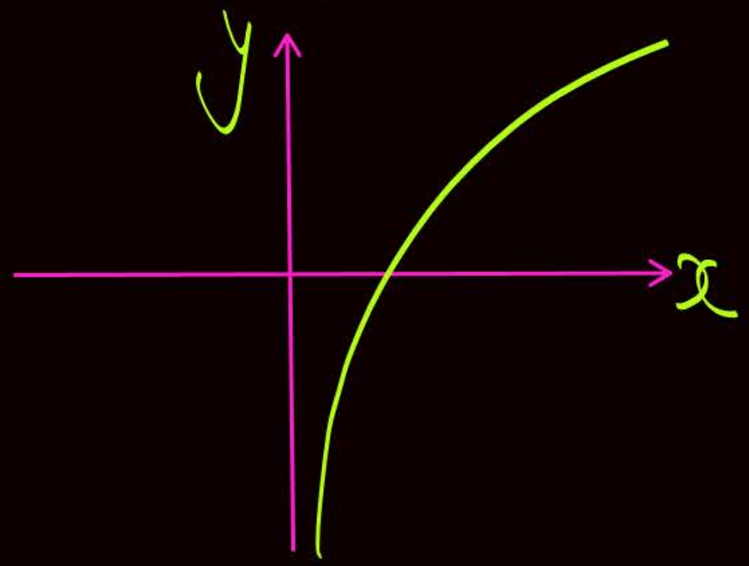
Q:- Check whether the functions 'f' and 'g' are identical or not ?

$f = \log(x^2)$

Here x^2 must be ">0" $\therefore$ Domain of $f = R - \{0\}$

$g = 2\log(x)$

Here x must be ">0" $\therefore$ Domain of $g = R^+$



Domain of $f \neq$ Domain of g

$\therefore f$ & g are not identical



Topic : Function composition

- Let 'f' and 'g' are two functions, Composite function of 'f' and 'g' (if exist) can be denoted by "fog"

In general, $f \circ g \neq g \circ f$ { function Composition is not Commutative }

- $(f \circ g)(x) = f(g(x))$
- $(g \circ f)(x) = g(f(x))$

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- $((f \circ g) \circ h)$

$((f \circ g) \circ h)(x)$

$(f \circ g)(h(x))$

$(f)(g(h(x)))$

$f(g(h(x)))$

$(f \circ (g \circ h))$

$(f \circ (g \circ h))$

$(f)((g \circ h)(x))$

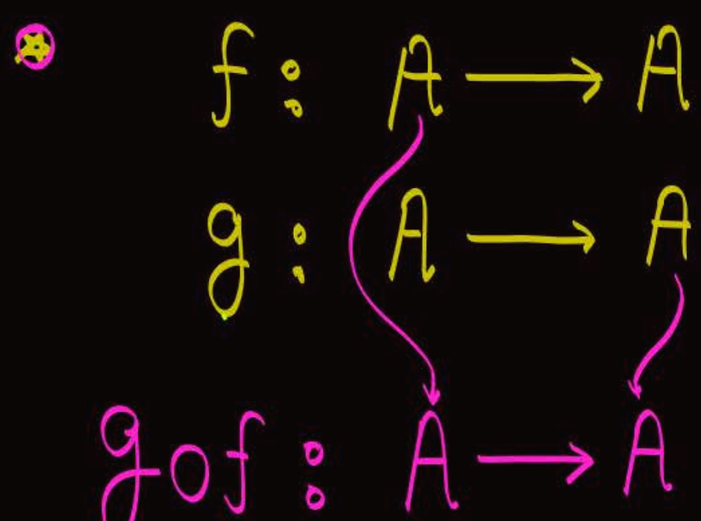
$(f)(g(h(x)))$

$f(g(h(x)))$

Equal

$\therefore (f \circ g) \circ h = f \circ (g \circ h)$

i.e., function composition is associative (Provided they are defined)

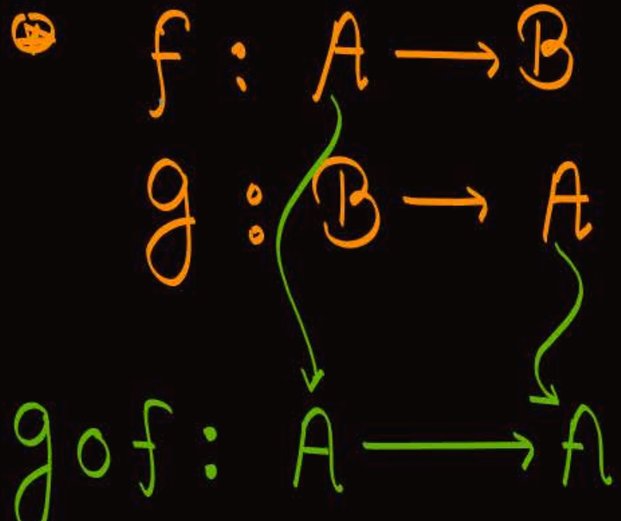


$\equiv$

$g(f(x))$: Domain of 'f' → Co-domain of g

fog : A → A

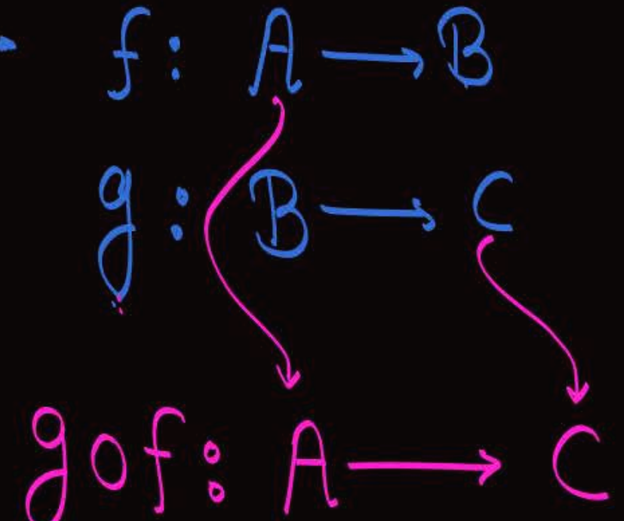
f(g(x)) : Domain of g → Codomain of f



g(f(x))

fog : B → B

f(g(x))



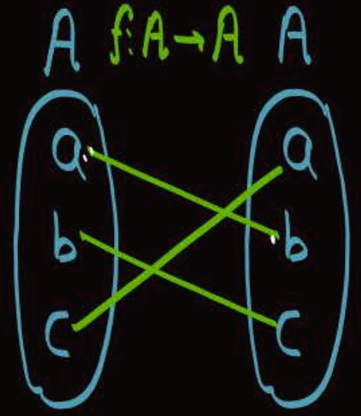
g(f(x))

fog : Need not be defined

$\Downarrow$

f(g(x)) : fog will be defined iff Range of g $\subseteq$ Domain of f

* Let $f: A \to A$ is a bijective function



$\therefore f^{-1}$ is also defined

$f^{-1}: A \to A$

$(f \circ f^{-1})(x) = x$

$f \circ f^{-1}: A \to A$

&

$f f^{-1} = I_A$

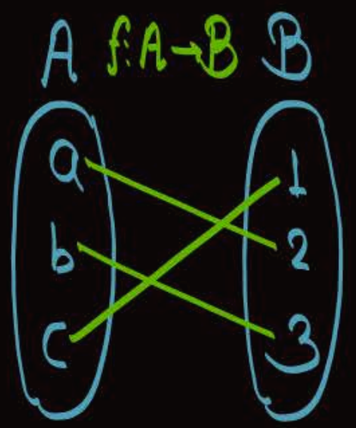
$(f^{-1} \circ f)(x) = x$

$f^{-1} \circ f: A \to A$

&

$f^{-1} f = I_A$

* Let $f: A \to B$ is a bijective function



$\therefore f^{-1}$ is also defined

$f^{-1}: B \to A$

$f^{-1} \circ f: A \to A$

$(f^{-1} \circ f)(x) = x$

$\therefore f^{-1} f = I_A$

$f \circ f^{-1}: B \to B$

$(f \circ f^{-1})(x) = x$

$f f^{-1} = I_B$

* Let $f: A \longrightarrow A$

$$f \circ I_A = f$$

$f \circ I_A : A \longrightarrow A$

$$(f \circ I_A)(x) = f(I_A(x)) = f(x) = f$$

$$I_A \circ f = f$$

$I_A \circ f : A \longrightarrow A$

$$(I_A \circ f)(x) = I_A(f(x)) = f(x) = f$$

it is a value from set A

Let $f: A \longrightarrow B$

$$f \circ I_A = f$$

$f \circ I_A : A \longrightarrow B$

$$(f \circ I_A)(x) = f(I_A(x)) = f(x) = f$$

$I_A \circ f$:

$I_A \circ f$: Need not be defined

$f \circ I_B$:

$f \circ I_B$: Need not be defined

$$I_B \circ f = f$$

$I_B \circ f : A \longrightarrow B$

$$(I_B \circ f)(x) = I_B(f(x)) = f(x)$$

Note :-

If '$*$' is any binary operation, then following property will always hold true

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

does not matter whether '$*$' is Commutative or not

$(a*b)^{-1} \neq a^{-1} * b^{-1}$, need not hold true in all situation.

We can say that $(a*b)^{-1} = a^{-1} * b^{-1}$ will true only if we already know that $*$ is Commutative

Note: We know that function Composition is not Commutative

$\therefore (f \circ g)^{-1} = g^{-1} \circ f^{-1}$ will always true

But $(fog)^{-1} = f^{-1} \circ g^{-1}$ need not hold true

H.W.

Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \ldots, n\}$ to $\{0,1\}$. Define the binary relation $\preccurlyeq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}$, $f \preccurlyeq g$ if and only if $\forall x \in \{1, \ldots, n\}$, $f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preccurlyeq$ is a symmetric relation
(B)	$(\mathcal{F}, \preccurlyeq)$ is a partial order
(C)	$(\mathcal{F}, \preccurlyeq)$ is a lattice
(D)	$\preccurlyeq$ is an equivalence relation

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H.W.

Q.17	$g(.)$ is a function from A to B, $f(.)$ is a function from B to C, and their composition defined as $f(g(.))$ is a mapping from A to C. If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?
(A)	$g(.)$ must be an onto (surjective) function.
(B)	$g(.)$ must be a one-to-one (injective) function.
(C)	$g(.)$ must be a bijective function, that is, both one-to-one and onto.
(D)	$g(.)$ is not required to be a one-to-one or onto function.

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HW Check whether the statement is True or False

① If fog is onto then 'f' must be onto

② If 'f' is onto then 'fog' is onto

③ If 'f' is one-one then fog is one-one

④ If 'fog' is one-one then 'f' must be one-one

⑤ If 'fog' is one-one, then 'g' must be one-one



2 mins Summary

Topic — Identity function and constant function

Topic — Identical function

Topic — Function composition

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THANK - YOU